Practice workbook · State the symptoms and effects of inherited conditions

Effects of polydactyly, haemophilia and cystic fibrosis on the body.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-STATE-INHERITED-CONDITION-EFFECTS

1. Question 6 · part b.ii · 4 marks

June 2025 Paper 2R

Worked-example retry · Attempt part b.ii only.

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6 (a) Individuals vary in their phenotypes.

State what is meant by the phenotype of an individual.

(1)

(b) (i) Cystic fibrosis is a disorder that affects epithelial membranes.

Describe what causes an individual to have cystic fibrosis.

(2)

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(ii) Describe the effects of cystic fibrosis on the health of an individual.

(4)

1. Question 6 · part b.ii · 4 marks

June 2025 Paper 2R

Worked-example retry · Attempt part b.ii only.

(iii) Explain why two parents who do not have cystic fibrosis can have a child who has cystic fibrosis.

(2)

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(Total for Question 6 = 9 marks)

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Working and answer

2. Question 3 · part a.iii · 1 mark

June 2024 Paper 1

Further practice · Attempt part a.iii only.

3 (a) Polydactyly is an inherited condition.

There are several faulty genes on different chromosomes that can cause polydactyly.

(i) State which part of a cell contains the most chromosomes.

(1)

(ii) Genes exist in alternative forms.

State the name given to an alternative form of a gene.

(1)

(iii) Give one symptom of polydactyly.

(1)

(b) Polydactyly is caused by the dominant form of the gene.

The table shows the genotypes of two parents, R and S.

Parent	Genotype
female (R)	Pp
male (S)	pp

(i) Name the term used to describe the genotype of parent R.

(1)

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2. Question 3 · part a.iii · 1 mark · Working and answer

Working and answer

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3. Question 9 · part b.ii · 1 mark

June 2024 Paper 1

Further practice · Attempt part b.ii only.

9 (a) There are four main blood groups.

One of these groups is known as type O.

(i) Name the other three blood groups.

(2)

(ii) A person with blood type O needs a blood transfusion.

Explain which blood type this person should receive in a transfusion.

(5)

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3. Question 9 · part b.ii · 1 mark

June 2024 Paper 1

Further practice · Attempt part b.ii only.

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(b) Haemophilia is a sex-linked genetic condition that affects the blood.

(i) Describe what is meant by a sex-linked genetic condition.

(2)

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(ii) State one effect that haemophilia will have on an individual.

(1)

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(Total for Question 9 = 10 marks)

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Working and answer

4. Question 4 · part c.ii · 1 mark

June 2024 Paper 2R

Further practice · Attempt part c.ii only.

4 (a) A student does a survey of eye colour in a sample of people.

The table shows a tally of the number of people with each eye colour in the sample.

Blue	Green	Brown	Hazel	Grey
𝍸 𝍸 𝍸 𝍸 𝍸	𝍸 𝍸 I	𝍸 𝍸 𝍸 𝍸 𝍸 𝍸 I	𝍸 𝍸 𝍸 𝍸 𝍸 𝍸	III

(i) Complete the table by giving the missing information.

(1)

Number of people with each eye colour				
Blue	**Green**	**Brown**	**Hazel**	**Grey**
25				3

(ii) Calculate the percentage of people with brown eyes in the sample.

(2)

percentage = ... %

(b) Eye colour is determined by alleles inherited from parents.

(i) Which of these gives the definition of an allele?

(1)

- ☒ **A** two dominant genes in a genotype
- ☒ **B** two recessive genes in a genotype
- ☒ **C** different genes in a genotype
- ☒ **D** alternative forms of the same gene in a genotype

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P 7 4 4 9 5 A 0 1 2 2 8

4. Question 4 · part c.ii · 1 mark

June 2024 Paper 2R

Further practice · Attempt part c.ii only.

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(ii) Explain why diploid cells contain two alleles for each characteristic.

(4)

(c) Cystic fibrosis is a condition that is also determined by alleles that are inherited from parents.

(i) Give the possible genotypes of two parents that have a child with cystic fibrosis.

(2)

(ii) Give one symptom of cystic fibrosis.

(1)

(Total for Question 4 = 11 marks)

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4. Question 4 · part c.ii · 1 mark · Working and answer

Working and answer

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